MADHAN KUMAR G

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Open to Remote | Willing to Relocate | Available for EST / GMT Timezone Overlap

SUMMARY

Senior Azure Data Engineer with **4.5 years** of experience in the full **Data Engineering Lifecycle** - scalable **Batch and Real-time pipelines, ETL/ELT, Data Warehousing, Big Data** and enterprise **Data Quality frameworks** on **Azure Databricks**. Expert in **PySpark, Delta Lake, Unity Catalog, ADF, Spark Structured Streaming** and CI/CD via **Databricks Asset Bundles (DABs)**.

CORE SKILLS

Azure Databricks, PySpark, Delta Lake, Azure Data Factory (ADF), ADLS Gen2, Spark Structured Streaming, Unity Catalog, Azure DevOps, CI/CD, SQL, Python, Big Data, Distributed Processing
Cloud & Storage: Azure Databricks, ADLS Gen2, Azure Data Factory (ADF), Azure Event Hubs, Blob Storage
Data Engineering: PySpark, Spark Structured Streaming, Delta Lake, Delta Live Tables, Batch Processing, Real-time Processing, Distributed Processing
Architecture: Medallion Architecture, Data Modeling, Data Warehousing, ETL/ELT, Star/Snowflake Schema, Scalable Pipelines, High-volume Data
Governance & CI/CD: Unity Catalog, Databricks Asset Bundles (DABs), Change Data Feed, Azure DevOps, Git
Tools: Azure Databricks, ADF, ADLS Gen2, Event Hubs, Azure DevOps, Lakeview Dashboards
Languages & DB: Python, SQL, MySQL, Azure SQL, Oracle, PostgreSQL
Certifications: DP-203: Azure Data Engineer Associate, AZ-900: Azure Fundamentals

PROFESSIONAL EXPERIENCE

Senior Data Engineer, NTT Data
Aug 2024 - Present | Hyderabad, India

Smart Monitoring - *Enterprise Data Quality Framework - Healthcare Domain*
- Architected **metadata-driven DQ framework** on Azure Databricks with **Unity Catalog** - 20+ modular PySpark notebooks across **11 functional layers** (orchestration, rules, evaluation, profiling, audit, notifications, CI/CD)
- Enabled **zero-code client onboarding** using **12 Delta config tables** with referential integrity (PKs, FKs, **VARIANT types**) - eliminating all hardcoded logic for scalable multi-tenant support
- Implemented **6 pluggable DQ rule types** (Null Check, Row Count, Unique Count, Date Spans, Termed/Void Members) producing **record-level violations** with full business key correlation
- Built **historical baseline threshold engine** computing asymmetric statistical bounds from rolling metric windows - enabling **automatic anomaly detection** on data quality and volume trends
- Created **Lakeview dashboard** with **SLA compliance** tracking (SLA_MET/NOT_MET/NOT_ARRIVED); deployed via **DABs** across dev/sit/prod using **Azure DevOps CI/CD**; parallel file dispatch via ThreadPoolExecutor

Spektra - *Real-time Streaming Platform - Multi-client, Cloud-native*
- Architected Bronze - Silver - Gold scalable streaming pipeline using **Spark Structured Streaming** on Databricks with Event Hubs ingestion, **Delta Lake** ACID storage, and fault-tolerant checkpointing for **high-volume data streams**
- Built metadata-driven multi-client processing via Delta config tables; achieved **30% reduction in streaming latency** through PySpark **distributed processing** optimizations; ingestion into **ADX** via Azure DevOps CI/CD

ITSM Analytics Pipeline - *Metadata-driven Batch ETL Pipeline - ServiceNow / Data Warehousing*
- Orchestrated ADF + Databricks medallion **ETL/ELT pipeline** across 7 ITSM modules; multi-source extraction from ServiceNow REST API, Oracle, PostgreSQL, SFTP into **ADLS Gen2**; applied **14+ DQ rules** and **Delta Merge** for incremental and full loads
- Implemented Reference Data Management in Silver layer for value standardization - improving reprocessing efficiency by **40%**; built Gold layer KPIs for **Power BI**; **Unity Catalog** for governance and lineage; **30% faster batch processing** via PySpark tuning

Digital Specialist Engineer, Infosys
Sep 2021 - Aug 2024 | Hyderabad, India

AMD Data Warehouse Modernization - *ETL Engineering - Semiconductor / High-Performance Computing*
- Built **scalable PySpark ETL pipelines** on Azure Databricks for **high-volume semiconductor data**; designed Delta Lake Star/Snowflake **Data Warehousing** models with FACT and Dimension tables; applied partitioning, caching, and **distributed processing** to meet SLA targets
- Implemented **Azure DevOps CI/CD** with feature/release branching across dev/staging/prod; applied wide and narrow Spark transformations for **Big Data batch processing** workloads

EDUCATION

Bachelor of Technology (B.Tech), Electronics and Communication Engineering
Sri Vidyanikethan Engineering College, Andhra Pradesh, India | 2021 | 77%